

Highlights

An allosteric metabolite glue couples AICAR inversely to one-carbon availability: a theoretical prediction and its therapeutic corollaries

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- A glue-regulated PPAT step replaces demand-slaved purine synthesis in the formate switch model.
- The model predicts AICAR falls as one-carbon availability rises — an inverse coupling.
- The inverse coupling requires the PPAT–NUDT5 glue; a NUDT5 knockout should reverse its sign.
- Calibration to binding data alone reproduces AICAR and PRPP data from two independent labs.
- Corollaries for AMPK signalling and thiopurine/antifolate cytotoxicity are derived.